

Can MLLMs Reason in Multimodality? EMMA: An Enhanced MultiModal ReAsoning Benchmark

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01 Motivation And Background

- Human reasoning integrates language and vision for spatial, physical, and diagrammatic tasks
- LLMs excel in text; MLLMs' true cross-modal reasoning remains unclear
- Existing benchmarks emphasize recall or shallow perception over integrated reasoning
- Gap: evaluating iterative visual-text interplay and multi-step multimodal reasoning

Multimodal reasoning question in EMMA

Question: Three point charges, of charge $+Q$, $-2Q$, and $+3Q$, are placed equidistant as shown. Which vector best describes the net direction of the electric force acting on the $+Q$ charge?

Human reasoning

The direction of the electric force due to $-2Q$ and $+3Q$ can be visualized with a quick free-body sketch of the electric forces:

Combining the forces:
 $\because |+3Q| > |-2Q|$
 Δ **repel** > **attract**
b. ✓

Model reasoning (GPT-4o)

1. Forces Due to the Charges:

- The $-2Q$ charge attracts the $+Q$ charge. (Force points downward-left).
- The $+3Q$ charge repels the $+Q$ charge. (Force points downward-right).

2. Force Magnitudes (Coulomb's Law):

- Force due to $-2Q$: $F_{-2Q} = \frac{2kQ^2}{r^2}$.
- Force due to $+3Q$: $F_{+3Q} = \frac{3kQ^2}{r^2}$.

Error: While the model understands that like charges repel, it incorrectly identifies the force from $+3Q$ to $+Q$ as **downward-right**, when it should be **upward-left**.

01 Limitations Of Current Benchmarks

Q: Is this an odd function? Yes or No	Q: A trip of the pupils to the zoo took 135 minutes. How many hours and minutes does it make?	Q: Chase wants to buy 4 kilograms of oval beads and 5 kilograms of star-shaped beads. How much will he spend?														
		<table border="1"> <tr><td>oval beads</td><td>\$2/kilogram</td></tr> <tr><td>rectangular beads</td><td>\$3/kilogram</td></tr> <tr><td>star-shaped beads</td><td>\$2/kilogram</td></tr> <tr><td>spherical beads</td><td>\$2/kilogram</td></tr> <tr><td>heart-shaped beads</td><td>\$3/kilogram</td></tr> <tr><td>square beads</td><td>\$2/kilogram</td></tr> <tr><td>flower-shaped beads</td><td>\$2/kilogram</td></tr> </table>	oval beads	\$2/kilogram	rectangular beads	\$3/kilogram	star-shaped beads	\$2/kilogram	spherical beads	\$2/kilogram	heart-shaped beads	\$3/kilogram	square beads	\$2/kilogram	flower-shaped beads	\$2/kilogram
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square beads	\$2/kilogram															
flower-shaped beads	\$2/kilogram															
C: The function is $y = 0.5^x$... ✓	C: The image contains a bus ... ✓	C: The image is a price list ... ✓														
C+Q: No ✓	C+Q: 2 h 15 min ✓	C+Q: \$18 ✓														
I+Q: No ✓	I+Q: 2 h 15 min ✓	I+Q: \$18 ✓														

02 MLLM Progress And Visual CoT

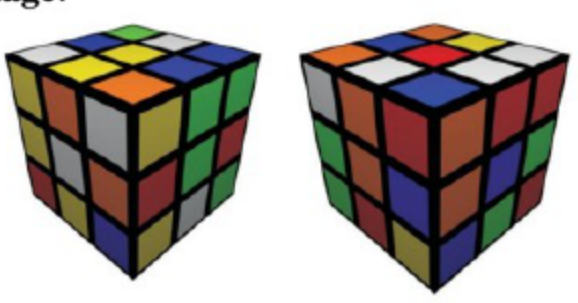
- MLLMs gain from visual instruction tuning, showing strong perception and some reasoning
- Visual CoT improves attention to regions/attributes but not visual manipulation or simulation
- Complex STEM tasks require cross-modal multi-hop reasoning beyond current methods

- Datasets allow language-only shortcuts or single visual pass solutions
- Filtering removes text-only items but leaves perception-aided, not reasoning-required problems
- Evaluations conflate perception with deeper visual simulation and integration

02 Benchmarks And Their Shortcomings

3D Spatial Simulation

Image:

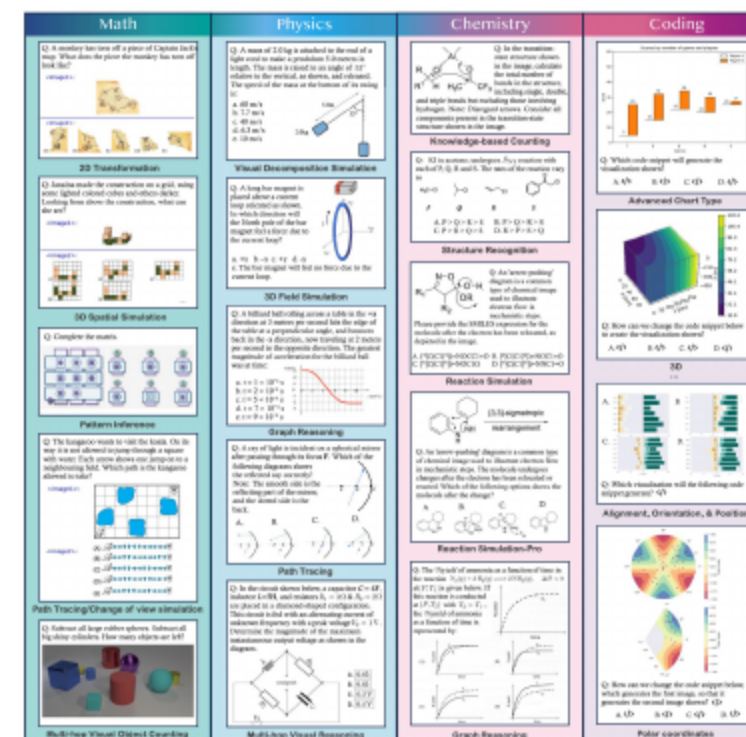


<image1>: <image2>:

Question: How many steps from the solution based on <image 1> and <image 2>?

- Redundancy between text and images enables bypassing visual reasoning
- MMMU-Pro improves filtering, but caption-plus-text can still solve items
- Need a benchmark requiring intrinsic visual reasoning and iterative cross-modal steps

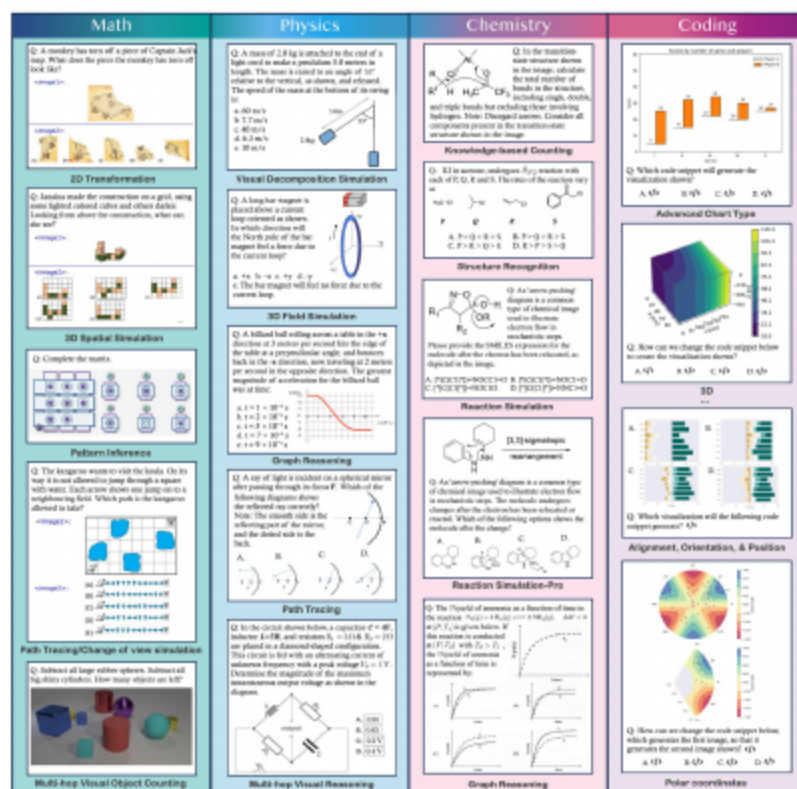
03 An Enhanced Multimodal Reasoning Benchmark



- EMMA: 2,788 problems across math, physics, chemistry, and coding
- Tasks unsolvable by text-only or single visual pass; require iterative cross-modal reasoning
- Concise answers enable reliable automated evaluation

03 Fine-Grained Skill Taxonomies And Labels

- Problems labeled for specific multimodal skills enabling granular diagnostics
- Covers patterns, transformations, decomposition, simulations, spatial visualization
- Expert-created/verified labels guide targeted capability research

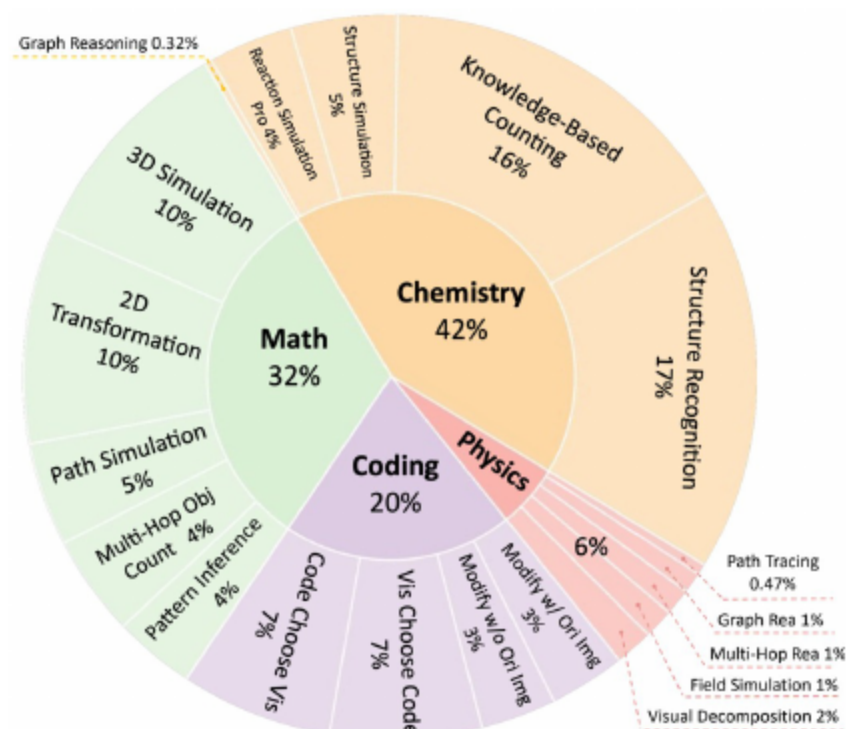


03 Stricter Filtering And Substantial New Data

Statistic	Number
Total questions	2,788
- Multiple-choice questions	2,002 (72%)
- Free-form questions	786 (28%)
- Questions with answers	2,788 (100%)
- Questions newly added	1,796 (64%)
Image in the question	2,599 (93%)
Image in the option(s)	195 (7%)
Problems with multiple images	298 (10%)

- Caption-augmented filter removes caption-plus-text solvable items
- Adds 1,796 new problems, expanding physics, chemistry, and coding
- Introduces standardized visualization-coding MCQ evaluation

04 Dataset Composition And Formats

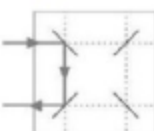


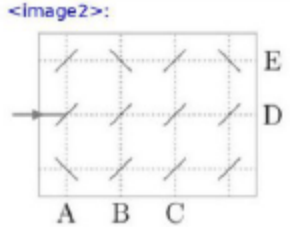
- 2,788 questions: 2,002 multiple-choice, 786 free-form
- Images in 93% of questions; 10% have multiple images
- EMMA-mini: 400 balanced questions for efficient comparisons

04 Skill Coverage Across Subjects

Path Tracing

Image:

<image1>: 

<image2>: 

Question: The two-sided mirrors reflect the laser beams as shown in the small picture: <image1>. At which letter does the laser beam leave the picture: <image2> ?

Change of View Simulation

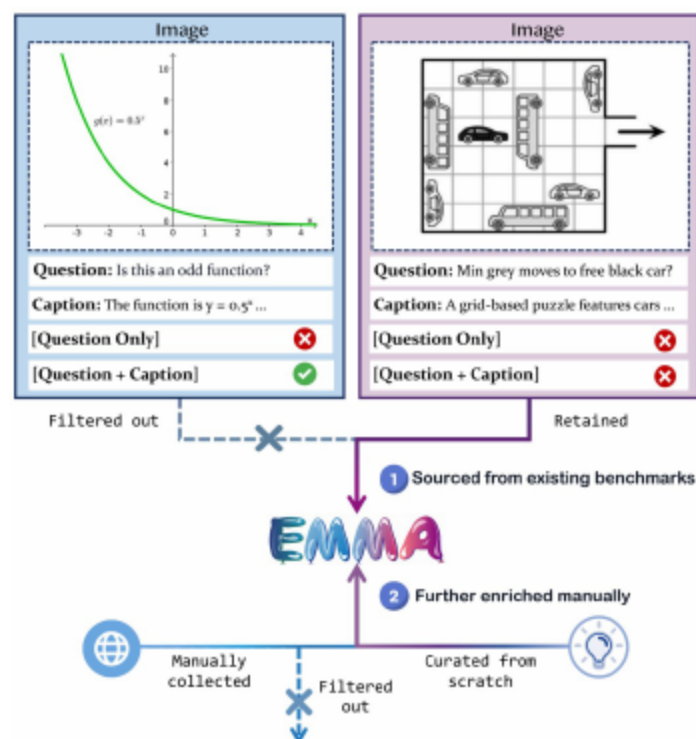
Image:



Question: In how many places in the picture are two children holding each other with their left hands?

- Math: 3D simulation, 2D transformation, path tracing, counting, pattern inference
- Physics: 3D field simulation, graph reasoning, path tracing
- Chemistry: structure recognition, bond counting, reaction simulation

05 Step 1: Source And Caption-Augmented Filtering



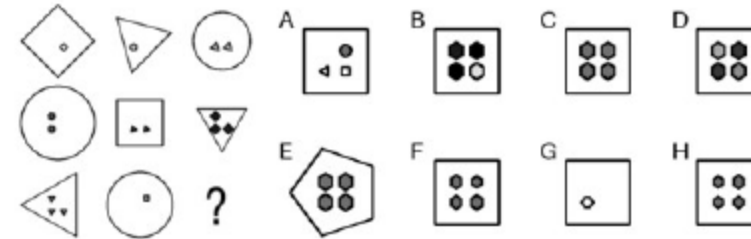
- Source from prior benchmarks and web materials
- Caption images; provide text plus captions to models
- Discard items any model solves $\geq 5/10$ across 30 trials
- Removes text-only and single-pass solvable problems

05 Step 2: Taxonomy Labeling And Expansion

- Categorize remaining items into fine-grained skill taxonomies
- Manually construct problems to cover underrepresented skills
- Chemistry via RDKit; reactions with expert annotations
- Coding seeded from curated visualizations with controlled variations

Source: RAVEN

Image:



Question: Choose the answer for the missing picture.

05 Subject-Specific Construction Details

- Math: refine MathVision/MathVista; add RAVEN for multi-hop reasoning
- Physics: aggregate OlympiadBench, EXAMS-V, MMMU, curated AP/Khan
- Chemistry: emphasize structures and reactions beyond formulas
- Coding: four visualization tasks via CharXiv and matplotlib exemplars

06 Data Splits And Human Baseline

- EMMA-mini: 100 balanced questions per subject for fast iteration
- Two experts per subject evaluate; average provides human baseline
- Benchmarks model gaps in multimodal reasoning

06 Models And Prompting Strategies

Model	Parameter Setting	Source	URL
GPT-4o pass@1	temperature = 0.0	chatgpt-4o-latest	https://platform.openai.com
GPT-4o pass@n	temperature = 0.7	chatgpt-4o-latest	https://platform.openai.com
Claude 3.5 Sonnet	temperature = 0.0	claude-3.5-sonnet	https://www.anthropic.com/
Gemini 2.0 Flash pass@1	temperature = 0.0	gemini-2.0-flash-exp	https://ai.google.dev/
Gemini 2.0 Flash pass@n	temperature = 0.7	gemini-2.0-flash-exp	https://ai.google.dev/
Gemini 2.0 Flash Thinking-1219	temperature = 0.0	gemini-2.0-flash-thinking-exp-1219	https://ai.google.dev/
Gemini 2.0 Flash Thinking-0121	temperature = 0.0	gemini-2.0-flash-thinking-exp-0121	https://ai.google.dev/
OpenAI o1	-	interface	https://chatgpt.com/
Qwen2-VL-72B-Instruct	temperature = 0.7	local checkpoint	https://huggingface.co/Qwen/Qwen2-VL-72B-Instruct
QVQ-72B-Preview	temperature = 0.7	local checkpoint	https://huggingface.co/Qwen/QVQ-72B-Preview
LLaVA-Onevision-72B	do_sample=True, temperature = 0.7	local checkpoint	https://huggingface.co/llava-hf/llava-onevision-qwen2-72b-ov-hf
InternVL2-Llama3-76B	do_sample=True, temperature = 0.7	local checkpoint	https://huggingface.co/OpenGVLab/InternVL2-Llama3-76B
InternVL2.5-78B	do_sample=True, temperature = 0.7	local checkpoint	https://huggingface.co/OpenGVLab/InternVL2_5-78B

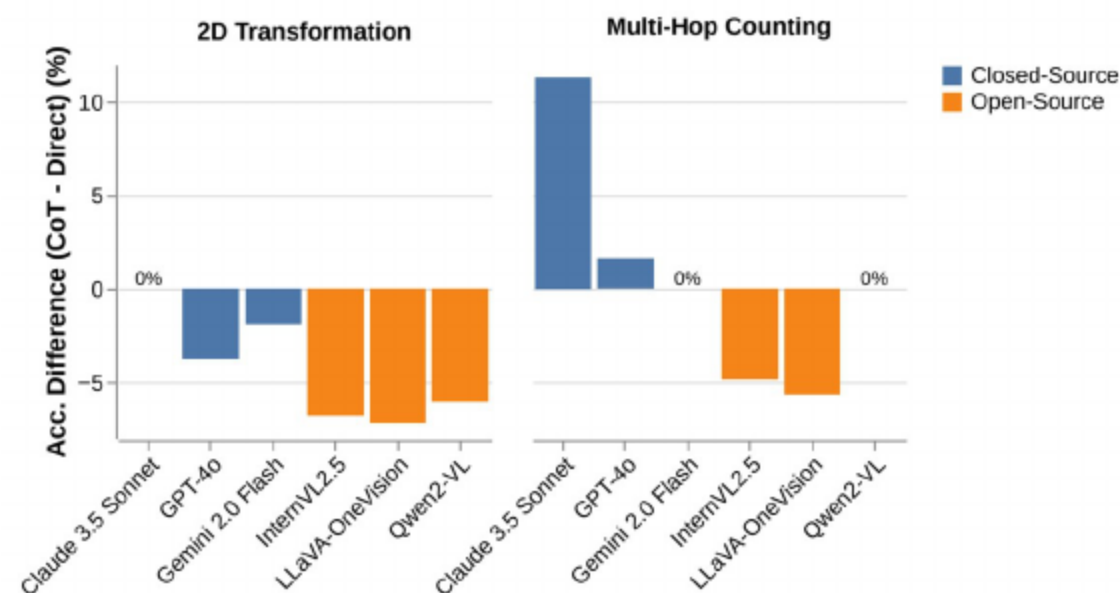
- Evaluate ten SoTA MLLMs: open-source and proprietary
- Two prompting modes: direct and Chain-of-Thought
- Assess impact of explicit textual reasoning

07 Overall Performance And Gaps To Humans

	CoT	EMMA					EMMA-mini				
		Math	Phys.	Chem.	Coding	Overall	Math	Phys.	Chem.	Coding	Overall
Random choice	—	(892)	(156)	(1,176)	(564)	(2,788)	(100)	(100)	(100)	(100)	(400)
Human Expert	—	14.01	25.64	16.50	25.71	18.08	13.00	23.00	27.00	28.00	22.75
Claude 3.5 Sonnet	✗	25.34	33.97	40.90	38.65	35.08	23.00	34.00	44.00	35.00	34.00
Gemini 2.0 Flash	✗	23.88	38.46	36.31	42.02	33.61	20.00	40.00	36.00	41.00	34.25
GPT-4o	✗	27.24	38.46	31.89	40.07	32.42	30.00	38.00	33.00	40.00	35.25
Qwen2-VL-72B-Instruct	✗	33.07	42.31	32.06	34.57	33.46	38.00	40.00	34.00	37.00	37.25
LLaVA-Onevision-72B	✗	27.69	35.90	25.26	28.72	27.33	25.00	32.00	24.00	28.00	27.25
InternVL2-Llama3-76B	✗	25.11	22.44	24.06	27.84	25.07	31.00	22.00	21.00	28.00	25.50
InternVL2.5-78B	✗	31.39	38.46	35.20	31.91	33.50	30.00	40.00	38.00	33.00	35.25
Claude 3.5 Sonnet	✓	29.37	41.03	41.07	40.60	37.23 (↑ 2.15)	30.00	38.00	41.00	39.00	37.00 (↑ 3.00)
Gemini 2.0 Flash	✓	25.90	38.46	24.66	40.96	29.12 (↓ 4.48)	24.00	41.00	36.00	44.00	36.25 (↑ 2.00)
GPT-4o	✓	25.56	43.59	33.67	39.01	32.71 (↑ 0.29)	27.00	44.00	35.00	38.00	36.00 (↑ 0.75)
Qwen2-VL-72B-Instruct	✓	27.69	34.62	24.57	29.43	27.12 (↓ 6.35)	35.00	34.00	32.00	23.00	31.00 (↓ 6.25)
LLaVA-Onevision-72B	✓	22.42	15.38	22.70	30.67	23.82 (↓ 3.52)	23.00	26.00	23.00	29.00	25.25 (↓ 2.00)
InternVL2-Llama3-76B	✓	22.20	32.05	19.73	30.32	23.35 (↓ 1.72)	27.00	33.00	21.00	32.00	28.25 (↑ 2.75)
InternVL2.5-78B	✓	25.56	39.74	27.47	25.18	27.08 (↓ 6.42)	31.00	36.00	24.00	19.00	27.50 (↓ 7.75)
Gemini 2.0 Flash Thinking-1219	—	31.61	56.41	37.93	43.44	38.06	35.00	57.00	41.00	41.00	43.50
Gemini 2.0 Flash Thinking-0121	—	37.11	60.26	41.58	48.05	42.50	34.00	63.00	47.00	48.00	48.00
QVQ-72B-Preview	—	—	—	—	—	—	34.00	39.00	24.00	31.00	32.00
o1	—	—	—	—	—	—	41.00	49.00	40.00	53.00	45.75

- All models underperform humans; best achieves 48.0% on EMMA-mini
- Human gap averages 29.75%; smallest in physics; large in math, chemistry, coding
- Closed-source generally lead; Gemini 2.0 Flash Thinking strongest

07 Effect Of Chain-Of-Thought Prompting



- Textual CoT helps many closed-source models
- Open-source models often degrade; >6% drops from direct prompting leaders
- CoT may induce hallucinations for tasks needing fine-grained visual simulation

08 Method 1: Majority Voting

Model	Method	Reward Model	N=1	N=2	N=4	N=8	N=16
GPT-4o	Majority Voting	—	—	—	37.25	36.25	38.25
	BoN	GPT-4o (Self)	—	35.50	35.75	36.75	—
	BoN	Gemini Flash Thinking	36.00	40.75	36.25	36.5	—
	Tournament	Gemini Flash Thinking	—	40.75	39.25	41.25	35.25
	Pass@N	—	—	45.00	53.25	65.75	74.00
Gemini 2.0 Flash	Majority Voting	—	—	—	37.75	39.25	39.75
	BoN	Gemini Flash (Self)	—	38.25	36.50	36.00	—
	BoN	Gemini Flash Thinking	36.25	36.75	37.00	40.25	—
	Tournament	Gemini Flash Thinking	—	36.75	37.25	40.75	38.75
	Pass@N	—	—	45.25	56.25	64.50	75.00
Gemini 2.0 Flash Thinking	Majority Voting	—	—	—	48.00	49.00	50.75
	Tournament	Gemini Flash Thinking (Self)	43.50	45.50	47.25	47.25	48.00
	Pass@N	—	—	53.75	64.50	71.50	81.50
ol	—	—	45.75	—	—	—	—

- Sample multiple CoT candidates; select most frequent answer
- Improves accuracy up to N=16 for stronger bases, still below humans
- Limited when visual reasoning steps are wrong or missing

08 Method 2: Best-Of-N With Reward Models

Model	Method	Reward Model	N=1	N=2	N=4	N=8	N=16
GPT-4o	Majority Voting	—	—	—	37.25	36.25	38.25
	BoN	GPT-4o (Self)	—	35.50	35.75	36.75	—
	BoN	Gemini Flash Thinking	36.00	40.75	36.25	36.5	—
	Tournament	Gemini Flash Thinking	—	40.75	39.25	41.25	35.25
	Pass@N	—	—	45.00	53.25	65.75	74.00
Gemini 2.0 Flash	Majority Voting	—	—	—	37.75	39.25	39.75
	BoN	Gemini Flash (Self)	—	38.25	36.50	36.00	—
	BoN	Gemini Flash Thinking	36.25	36.75	37.00	40.25	—
	Tournament	Gemini Flash Thinking	—	36.75	37.25	40.75	38.75
	Pass@N	—	—	45.25	56.25	64.50	75.00
Gemini 2.0 Flash Thinking	Majority Voting	—	—	—	48.00	49.00	50.75
	Tournament	Gemini Flash Thinking (Self)	43.50	45.50	47.25	47.25	48.00
	Pass@N	—	—	53.75	64.50	71.50	81.50
ol	—	—	45.75	—	—	—	—

- Select best among N using reward model; stronger external models work better
- Self-rewarding underperforms; improvements modest overall
- Reward models struggle with complex multimodal reasoning assessment

08 Method 3: Tournament Selection

Model	Method	Reward Model	N=1	N=2	N=4	N=8	N=16
GPT-4o	Majority Voting	—	—	—	37.25	36.25	38.25
	BoN	GPT-4o (Self)	—	35.50	35.75	36.75	—
	BoN	Gemini Flash Thinking	36.00	40.75	36.25	36.5	—
	Tournament	Gemini Flash Thinking	—	40.75	39.25	41.25	35.25
	Pass@N	—	—	45.00	53.25	65.75	74.00
Gemini 2.0 Flash	Majority Voting	—	—	—	37.75	39.25	39.75
	BoN	Gemini Flash (Self)	—	38.25	36.50	36.00	—
	BoN	Gemini Flash Thinking	36.25	36.75	37.00	40.25	—
	Tournament	Gemini Flash Thinking	—	36.75	37.25	40.75	38.75
	Pass@N	—	—	45.25	56.25	64.50	75.00
Gemini 2.0 Flash Thinking	Majority Voting	—	—	—	48.00	49.00	50.75
	Tournament	Gemini Flash Thinking (Self)	43.50	45.50	47.25	47.25	48.00
	Pass@N	—	—	53.75	64.50	71.50	81.50
ol	—	—	45.75	—	—	—	—

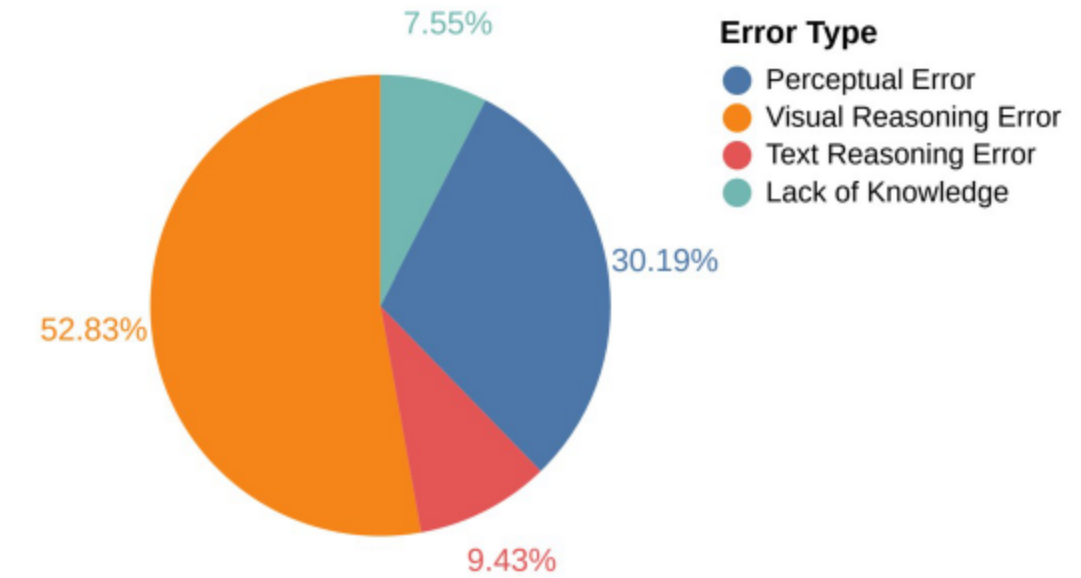
- Pairwise competitions adjudicated by strong reward model
- Outperforms Best-of-N via comparative judgments
- Still leaves sizable gap to human experts

08 Scaling Insights And Upper Bounds

Model	Method	Reward Model	N=1	N=2	N=4	N=8	N=16
GPT-4o	Majority Voting	—	—	—	37.25	36.25	38.25
	BoN	GPT-4o (Self)	—	35.50	35.75	36.75	—
	BoN	Gemini Flash Thinking	36.00	40.75	36.25	36.5	—
	Tournament	Gemini Flash Thinking	—	40.75	39.25	41.25	35.25
Gemini 2.0 Flash	Pass@N	—	—	45.00	53.25	65.75	74.00
	Majority Voting	—	—	—	37.75	39.25	39.75
	BoN	Gemini Flash (Self)	—	38.25	36.50	36.00	—
	BoN	Gemini Flash Thinking	36.25	36.75	37.00	40.25	—
Gemini 2.0 Flash Thinking	Tournament	Gemini Flash Thinking	—	36.75	37.25	40.75	38.75
	Pass@N	—	—	45.25	56.25	64.50	75.00
	Majority Voting	—	—	—	48.00	49.00	50.75
	Tournament	Gemini Flash Thinking (Self)	43.50	45.50	47.25	47.25	48.00
o1	Pass@N	—	—	53.75	64.50	71.50	81.50
	—	—	45.75	—	—	—	—

- Pass@N increases with stronger base reasoning; coverage improves
- Beyond N≈8, weaker models degrade
- Better base and reward models raise ceilings but cannot replace visual simulation

09 Visual Reasoning As The Bottleneck



- Errors concentrate in tasks needing precise spatial simulation and integration
- For o1 on EMMA-mini math/coding, visual reasoning errors exceed 50%
- Surpass perceptual, textual reasoning, and knowledge errors

09 When Textual CoT Helps Or Hurts

Math: 2D Transformation

Question: Rebecca folds a square piece of paper twice. Then she cuts off one corner as you can see in the diagram. Then she unfolds the paper. What could the paper look like now?

GPT-4o with CoT: To solve this problem, we need to carefully analyze the folding and cutting process and determine the resulting pattern when the paper is unfolded. ✓ Step 1: Understand the folding ✓ Step 2: Understand the cutting process

- Rebecca cuts off one corner of the folded paper. ✓
 - Since the paper is folded into a smaller square, this "corner" is actually a corner of the folded square, which corresponds to multiple corners of the original square. ✗

Final Answer: \boxed{E}

Error Analysis: After folding into 1/4, the "corners" of the folded paper are not the original corners. The solution fails to properly simulate the unfolding process.

Direct GPT-4o: B. ✓

Human Reasoning (Ground Truth):

- CoT hurts visually centric tasks like 2D transformations
- Helps tasks amenable to linguistic scaffolding, e.g., multi-hop counting
- Open-source models show increased hallucinations with CoT

09 Representative Failure Cases

- Models recall principles but mis-simulate spatial directionality
- Confusions in pattern/field tasks and skipped graphical decomposition
- Coding: miss subtle styling or scale differences

Physics: 3D Field Simulation

Question: A long bar magnet is placed above a current loop oriented as shown. In which direction will the North pole of the bar magnet feel a force due to the current loop?

a. +x b. -x c. +y d. -y e. The bar magnet will feel no force due to the current loop.

o1: A current loop behaves like a small bar magnet whose magnetic-dipole moment points according to the Right-Hand Rule. From the figure, the current goes around the loop in such a way that if you curl your right-hand fingers in the direction of the current, your thumb points up (the +y direction). Consequently, the loop acts like a magnet whose "north" side is on top (at +y). Hence the force on the north pole is c. +y.

Error Analysis: The error occurred because of limited visual spatial imagination, leading to the wrong conclusion about the direction of the thumb when applying the Right-Hand Rule. Additionally, the 3D spatial nature of the magnetic field is not properly simulated, leading to a faulty reasoning of the "north" pole's position.

Human Reasoning (Ground Truth): The current loop will produce a magnetic field B according to the Right-Hand Rule as shown below.

Then, as shown on the left, B points out from the middle of the current loop I , directed in the +x direction at the location where the bar magnet is located. This magnetic field, pointing to the right, will have the effect of repelling the North pole of the bar magnet, forcing the bar magnet to the right, in the +x direction. The correct answer is a.

Conclusions And Takeaways

EMMA isolates genuine multimodal reasoning by eliminating shortcuts

Large human gaps; limited gains from CoT and scaling

Visual reasoning is the dominant bottleneck to address

Benchmark Limitations

Manual curation and balance may introduce biases; physics underrepresented

Future models may find new shortcuts; requires periodic refresh

Broaden chemistry topics and diversify sources for robustness

- Architectures enabling visual simulation, spatial manipulation, grounded visual CoT
- Richer supervision, diagram tool-use, integrated reward modeling
- EMMA sets a higher standard toward trustworthy multimodal reasoning



THANKS!